

How do Arrays Decay into Pointers?

This article is for C programming newbies and enthusiasts. It deals with how arrays are passed as parameters to a function decays into pointers.



Many newbies to the C world are often confused about passing an array as the parameter to a function. They find it difficult to decide whether it is done using the 'pass by value' or 'pass by reference' parameter passing mechanism. This article explains how the arrays are passed to the function and why this is so.

Example 1: 'An array name as a function parameter is a pointer'

In this example, consider 1-D array to explain the meaning of the above statement with the help of the demo codes. Consider Code 1, which contains a function called `display()` to display the elements of the array.

```
1 #include <stdio.h>
2
3 void display(int numbers[], int size);
4
5 int main()
6 {
7     //Definition of an array
8     int numbers[] = {10, 20, 30, 40, 50};
9
10    //Calling display function, to print the elements
11    display(numbers, sizeof(numbers) /
12    sizeof(numbers[0]));
13
14    return 0;
15 }
16 //Function definition
17 void display(int numbers[], int size)
18 {
19     int i;
```

```
20     for (i = 0; i < size; i++)
21     {
22         printf("The value @ address: %p:
23         %d\n", &numbers[i], numbers[i]);
24     }
```

Code 1: This code is to demonstrate how arrays are passed to functions

From Code 1, whenever arrays are passed as the arguments to functions, they are always passed by using the 'Pass by reference' mechanism. Because of this, they will decay into pointers in the function parameters. In Line 17 of Code 1, In main function, the name of array (numbers) which contains the base address of the array (numbers) is passed to the function called the display, and collected using the parameter called numbers, which is nothing but the pointer to the base address of the array (numbers) defined in the main function. The function `display()` is equivalent to the code shown below.

```
void display(int *numbers, int size);
```

The definition of the display is also given in Code 2.

```
1 //Function definition
2 void display(int *numbers, int size)
3 {
4     int i;
5     for (i = 0; i < size; i++)
6     {
7         printf("The value @ address: %p: %d\n", &numbers[i],
8         numbers[i]);
9     }
```

Code 2